torch.bincount#

https://pytorch.org/docs/stable/generated/torch.bincount.html#torch.bincount

Description:

Count the frequency of each value in an array of non-negative ints. The number of bins (size 1) is one larger than the largest value in input unless input is empty, in which case the result is a tensor of size 0. If minlength is specified, the number of bins is at least minlength and if input is empty, then the result is tensor of size minlength filled with zeros. If n is the value at position i, out[n] += weights[i] if weights is specified else out[n] += 1. This operation may produce nondeterministic gradients when given tensors on a CUDA device. See Reproducibility for more information. input (Tensor) – 1-d int tensor

Function Signature

```
torch.bincount(input, weights=None, minlength=0) → Tensor#
Parameters
Returns
Return type
```

Returns:

output (Tensor)

Code Examples

```
>>> input = torch.randint(0, 8, (5,), dtype=torch.int64)
>>> weights = torch.linspace(0, 1, steps=5)
>>> input, weights
(tensor([4, 3, 6, 3, 4]),
 tensor([ 0.0000,  0.2500,  0.5000,  0.7500,  1.0000])

>>> torch.bincount(input)
tensor([0, 0, 0, 2, 2, 0, 1])

>>> input.bincount(weights)
tensor([0.0000, 0.0000, 0.0000, 1.0000, 1.0000, 0.0000, 0.5000])
```

Notes & Warnings

NOTE Note This operation may produce nondeterministic gradients when given tensors on a CUDA device. See Reproducibility for more information.

Detailed Documentation

Documentation

Count the frequency of each value in an array of non-negative ints.

The number of bins (size 1) is one larger than the largest value in input unless input is empty, in which case the result is a tensor of size 0. If minlength is specified, the number of bins is at least minlength and if input is empty, then the result is tensor of size minlength filled with zeros. If n is the value at position i, out[n] += weights[i] if weights is specified else out[n] += 1.

This operation may produce nondeterministic gradients when given tensors on a CUDA device. See Reproducibility for more information.

input (Tensor) – 1-d int tensor

weights (Tensor) – optional, weight for each value in the input tensor. Should be of same size as input tensor.

minlength (int) – optional, minimum number of bins. Should be non-negative.

a tensor of shape Size([max(input) + 1]) if input is non-empty, else Size(0)